DAMAN ARORA

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EDUCATION

Carnegie Mellon University 2024–2029 (expected)
Ph.D. in Machine Learning; GPA: **4.0**
Advisor: Andrea Zanette

Indian Institute of Technology Delhi 2018–2023
B.Tech. and M.Tech. in Computer Science; CGPA: **9.4**
Thesis: Learning Generalized Neural Policies for Relational MDPs
Advisors: Mausam and Parag Singla

PUBLICATIONS (* denotes equal contribution)

Xuecheng Liu*, **Daman Arora***, Gokul Swamy, Andrea Zanette, "Message Passing Enables Efficient Reasoning," *Conference on Language Modeling (COLM),* **Outstanding Paper**, 2026. arXiv:2607.01077.

Daman Arora, Andrea Zanette, "Training Language Models to Reason Efficiently," *Conference on Neural Information Processing Systems (NeurIPS)*, 2025. arXiv:2502.04463.

Fahim Tajwar*, Guanning Zeng*, Yueer Zhou, Yuda Song, **Daman Arora**, Yiding Jiang, Jeff Schneider, Ruslan Salakhutdinov, Haiwen Feng, Andrea Zanette, "Maximum Likelihood Reinforcement Learning," *International Conference on Machine Learning (ICML),* **Oral***; Best Paper at SPOT (ICLR)*, 2026. arXiv:2602.02710.

Daman Arora*, Himanshu Gaurav Singh*, Mausam, "Have LLMs Advanced Enough? A Challenging Problem Solving Benchmark for Large Language Models," *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023. arXiv:2305.15074.

Guanning Zeng*, Zhaoyi Zhou, **Daman Arora**, Andrea Zanette, "Shrinking the Variance: Shrinkage Baselines for Reinforcement Learning with Verifiable Rewards," *International Conference on Machine Learning (ICML)*, 2026. arXiv:2511.03710.

Vishal Sharma*, **Daman Arora***, Mausam, Parag Singla, "SymNet 3.0: Exploiting Long-Range Influences in Learning Generalized Neural Policies for Relational MDPs," *Conference on Uncertainty in Artificial Intelligence (UAI)*, 2023. PMLR.

Vishal Sharma, **Daman Arora**, Florian Geißer, Mausam, Parag Singla, "SymNet 2.0: Effectively Handling Non-Fluents and Actions in Generalized Neural Policies for RDDL Relational MDPs," *Conference on Uncertainty in Artificial Intelligence (UAI)*, 2022. PMLR.

WORKSHOP PUBLICATIONS

Daman Arora*, Nalin Wadhwa*, Atharv Sonwane*, Abhav Mehrotra, Saiteja Utpala, Ramakrishna B Bairi, Aditya Kanade, Nagarajan Natarajan, "MASAI: Modular Architecture for Software-engineering AI Agents," *NeurIPS Workshop on Open-World Agents*, 2024. OpenReview.

Daman Arora, Subbarao Kambhampati, "Learning and Leveraging Verifiers to Improve Planning Capabilities of Pre-trained Language Models," *ICML Workshop on Knowledge and Logical Reasoning in the Era of Data-driven Learning*, 2023. arXiv:2305.17077.

PREPRINTS

Daman Arora, Xuecheng Liu, Himanshu Gaurav Singh, Fahim Tajwar, Abitha Thankaraj, Ruslan Salakhutdinov, Haiwen Feng, Yuda Song, Andrea Zanette, "Training Language Models to Fail Gracefully," *arXiv preprint*, 2026. arXiv:2610.20401.

Alvin Zhang, Xuecheng Liu, Zixuan Wang, Fahim Tajwar, **Daman Arora**, Ruslan Salakhutdinov, Daniel Khashabi, Yuda Song, Andrea Zanette, "Harness Learning Enables Generalizable Test-Time Adaptation," *arXiv preprint*, 2026. arXiv:2609.35738.

Shrinivas Ramasubramanian, **Daman Arora**, Fahim Tajwar, Guanning Zeng, Qingyang Wu, Zhongzhu Zhou, Chenfeng Xu, Haiwen Feng, Yuda Song, Aarti Singh, Ruslan Salakhutdinov, J. Andrew Bagnell, Jeff Schneider, Andrea Zanette, "Tail-Likelihood Reinforcement Learning," *arXiv preprint*, 2026. arXiv:2609.02987.

Daman Arora*, Anush Kini*, Sayak Ray Chowdhury, Nagarajan Natarajan, Gaurav Sinha, Amit Sharma, "GAR-meets-RAG Paradigm for Zero-Shot Information Retrieval," *arXiv preprint*, 2023. arXiv:2310.20158.

HONORS

Outstanding Paper, Conference on Language Modeling	2026
IIT Delhi Merit Award (top 7%)	2019
KVPY All-India Rank 15	2018
JEE Advanced All-India Rank 158; JEE Main All-India Rank 118	2018

WORK EXPERIENCE

Microsoft Research India June 2023–June 2024
Research Fellow
Host: Nagarajan Natarajan; Retrieval and Software Engineering Agents

Sprinklr May 2022–July 2022
Software Engineer Intern

UiPath May 2021–July 2021
Software Engineer Intern